

Object Detection Extension

<https://github.com/aiyshawy/Object-Detection-Extension-in-Rapid-Miner>

[Python](#)

[PyTorch](#)

[ONNX Runtime](#)

[RapidMiner](#)

[Object Detection](#)

OVERVIEW

A RapidMiner Studio extension that brings transfer learning-based object detection into a no-code environment, supporting fine-tuning and inference with Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet models.

IMPACT

Developed two RapidMiner operators for object detection — `FineTuneObjectDetectionModel` and `ObjectDetectionModelInference` — enabling users to train and deploy detection models within a visual workflow without writing code.

TECHNICAL WALKTHROUGH

Object Detection Extension

Problem Statement

Different object detection use cases require different trade-offs between accuracy, speed, and computational cost. A single model cannot satisfy all scenarios.

Approach

Developed a unified object detection extension using PyTorch, supporting

multiple architectures

Faster R-CNN

FCOS

SSD

SSD Lite

RetinaNet

YOLO (extended work)

Training

Dataset annotated using XML bounding boxes

Transfer learning using COCO pre-trained weights

Fully configurable training pipeline:

Model selection

Batch size

Epochs

Learning rate

Momentum

Weight decay

Evaluation metrics:

IoU (Intersection over Union)

Precision

Recall (per class)

Inference

Users can choose any trained or pre-trained model

Output includes:

Image with bounding boxes, labels, and confidence scores

Tabular output with detected objects

Results also generated in base64 format for easy deployment and integration

Key Observations**Faster R-CNN performed best for accuracy**

SSD and SSD Lite were faster but had lower recall

RetinaNet handled class imbalance well**Model choice depends on application requirements****KEY DETAILS**

- FineTuneObjectDetectionModel accepts image and annotation directories (YOLO/Pascal VOC) and outputs model weights and class list CSV.
- ObjectDetectionModelInference runs predictions on new images and returns bounding boxes and class labels as a DataFrame.
- Supports five architectures: Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet for varying speed-accuracy trade-offs.
- Configurable for CPU-only and GPU (CUDA cu118) environments; installed as a JAR extension in RapidMiner Studio.
- Enables domain-specific object detection with minimal ML expertise using a drag-and-drop process designer.